

RESISTANCE TO EXTINCTION OF A LEVER PRESS IN SPINY MICE (*ACOMYS CAHIRINUS*)

RICHARD DENI

*John F. Kennedy Center for Research
on Education and Human Development¹*

Summary.—The operant lever pressing of 12 adult male spiny mice (*Acomys cahirinus*) was assessed during two phases of procedure: (1) CRF performance and subsequent extinction and (2) FR-10 performance and subsequent extinction. This highly precocial species showed a marked inability to inhibit previously reinforced responding during extinction following CRF performance. This result parallels extinction responding previously observed in guinea pigs, also an early developing species. Introduction of an FR-10 schedule resulted in a significant increase in lever pressing during performance sessions, establishing a contingency between responding and reinforcement that may not have existed during phase 1. However, complete extinction following FR-10 performance did not occur.

The spiny mouse (*Acomys cahirinus*) is a desert rodent possessing well developed sensory-motor capacities at birth (Ruch, 1967). The development of behavior in this animal has begun to be studied (Porter & Etscorn, 1976; Porter, Deni, & Doane, in press). With respect to studies of conditioning, these animals do not seem to be able to inhibit previously learned responses. Etscorn (1977) found adult male spiny mice took up to 36 daily extinction sessions before adequately reducing their performance of a previously learned avoidance response. In this respect spiny mice have demonstrated behavior similar to other precocial rodents. In a comparison of repeated operant acquisition and extinction in rats and guinea pigs, Braveman (1973) observed that while rats quickly inhibited responding during extinction, guinea pigs did not. Response rate during extinction for guinea pigs was highly variable within a range of 25 to 55 responses per 10-min. session. Responding during extinction for guinea pigs was significantly higher when compared to that for rats.

Braveman used a continuous reinforcement schedule (CRF) in that study. It is unclear to what extent the behavior of the guinea pigs was under the control of the food reinforcement since they did not show extinction when reinforcement was terminated. The present study sought to examine CRF and fixed-ratio (FR) operant performance and extinction in spiny mice. The FR phase of procedure was designed to assess the degree to which operant responding was under the control of the reinforcer. If resistance to extinction does occur and response rate cannot be decreased by removing reinforcement, it will be important to know whether response rate can be increased by requiring more responses to achieve reinforcement.

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METHOD

Subjects

Subjects were 12 male spiny mice (*Acomys cahirinus*), 40 to 50 days old at the start of the study. They were housed individually in plastic mouse cages with stainless steel wire covers. Their diet consisted of Purina mouse chow, supplemented with fresh carrot and apple.

Apparatus

Two identical Grason Stadler No. 1111 operant chambers were used. Each chamber was located inside a Grason Stadler No. 1101 sound-attenuating research chest. Two rat levers were located on the intelligence panel of each chamber. A metal extension was attached to each lever to facilitate operation by small rodents. A solenoid operated dipper delivered 0.1 ml of apple juice as liquid reinforcement for lever pressing. The dipper remained in the access position for 8 sec. with each operation. White noise was provided to each chamber through a small speaker located on the intelligence panel. Intensity was set at 80 dB measured at the speaker grill. A single 28-v DC lamp illuminated the chamber during testing. Solid state and electromechanical equipment was used to program contingencies and record responses.

Procedure

Phase 1—CRF.—Subjects were habituated to the operant chamber 30 min. per day for five consecutive days. During these sessions the levers did not operate to produce reinforcement, subjects were not deprived, and the dipper delivered a free reinforcement once every 60 sec. Following habituation CRF lever pressing was assessed for 8 consecutive daily 30-min. sessions. During these sessions both levers inside each chamber were operative. Total lever pressing was recorded. The only source of moisture for these animals was that obtained during the operant sessions. They were fed dry food in their home cages. Following Session 8, subjects continued daily 30-min. sessions under extinction conditions. Extinction lasted for 24 additional daily sessions. During these sessions the reinforcement dipper would operate but no apple juice was present. Following each extinction session, animals received 5 gm of fresh apple as their daily moisture provision.

Phase 2—FR-10.—Immediately following Phase 1, subjects were further tested during 5 daily 30-min. sessions where levers were again operative, and apple juice was given on an FR-10 schedule. No additional moisture supplements were given during these 5 sessions. Following FR-10 performance a second series of 12 extinction sessions occurred. Conditions were identical to extinction conditions in Phase 1.

RESULTS AND DISCUSSION

The mean number of presses for this group of *Acomys* during the last habituation session was 10.4 ($SD = 4.1$). Fig. 1 shows the mean of total lever presses per session for the 12 animals during CRF performance and extinction. Note that *Acomys* quickly exhibited a stable record of responding during early sessions. The mean number of responses earned apple juice in an amount roughly equivalent to average daily water consumption in adult males of this species (Etscorn, 1977). During extinction, however, lever pressing showed no tendency to diminish but maintained the pattern established for CRF performance. Mean lever presses by each animal were obtained for performance and extinction, and a correlated t test was used to determine the difference between

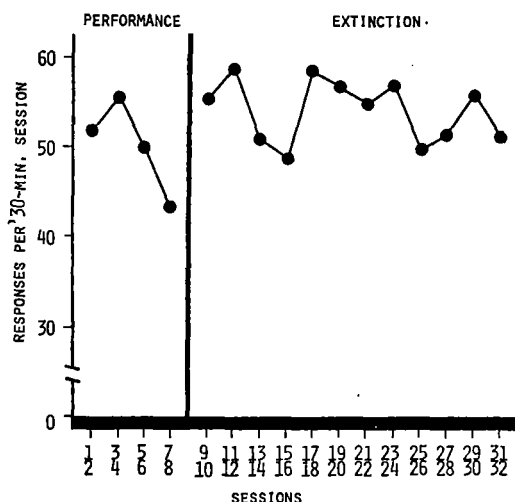


FIG. 1. Mean lever presses during CRF performance and extinction following CRF sessions

them. Table 1 shows these data and test results. There was no difference in over-all responding between performance and extinction conditions.

Fig. 2 shows FR-10 performance and extinction results. Responding increased significantly over CRF (for correlated data, $t = 10.16$, $df = 11$, $p < .001$), indicating that lever pressing was under the control of the apple juice reinforcement. Extinction following FR-10 performance shows the characteristic slope of an extinction curve and response mean was significantly lower than the mean for performance sessions (Table 1). Yet responding did not

TABLE 1
MEANS, STANDARD DEVIATIONS, AND t TESTS FOR LEVER PRESS PERFORMANCE AND EXTINCTION DURING CRF AND FR-10 SCHEDULES

Schedule		Performance	Extinction	t_r	p
CRF	M	50.0	53.9	0.56	<.001
	SD	10.8	20.2		
FR-10	M	215.6	93.4	11.83	
	SD	54.2	29.6		

drop below 60 presses per session. Extinction following CRF and extinction following FR-10 performance are similar once steady state has been reached, yet over-all means show responding during Phase 2 extinction to be greater (for correlated data, $t = 8.99$, $df = 11$, $p < .001$).

These findings of resistance to extinction following CRF and FR performance in the highly precocial *Acomys* are compatible with Braveman's (1973) re-

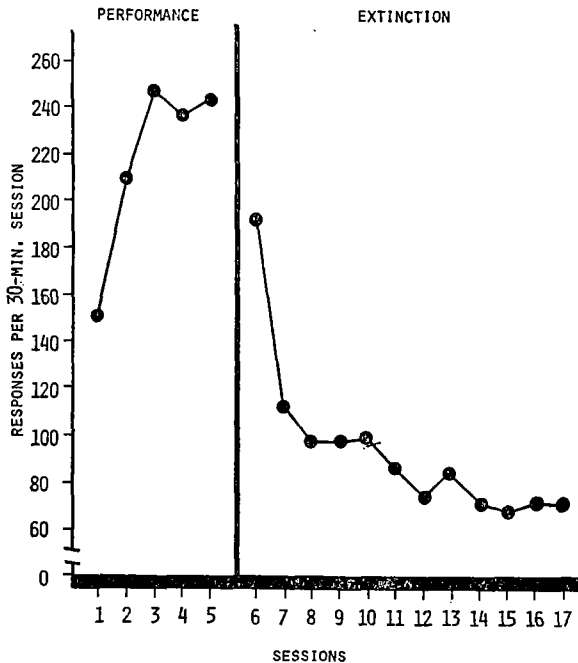


FIG. 2. Mean lever presses during FR-10 performance and the following extinction sessions

port of failure of guinea pigs to extinguish an operant. The significant increase in responding by *Acomys* during FR-10 verifies that these animals were performing under a reinforcement contingency. Braveman's conclusion that precocial species are less able to inhibit responding when compared to altricial species is supported by these findings with *Acomys*. This extinction differential may however be present as a function of developmental level especially pronounced for the young adult animals. It would be interesting to note the course of the differential throughout the life span of *Acomys* and other precocial mammals.

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